# Chapter 1: Introduction

Agricultural waste management is one of the most critical environmental challenges facing India today. Every year, approximately 500 million tonnes of crop residue is generated, out of which a significant portion is burned in open fields. This practice leads to severe air pollution, soil degradation, loss of nutrients, and contributes significantly to greenhouse gas emissions. The National Capital Region (NCR) witnesses hazardous air quality levels every winter, primarily due to stubble burning in neighboring states.

Despite the environmental and health hazards, farmers often resort to burning because they lack accessible, affordable, and convenient alternatives for waste disposal. There is a disconnect between farmers who generate waste and the recycling industries that can convert this waste into valuable products such as biofuel, compost, animal feed, and building materials.

AgriTrace bridges this gap by providing a digital platform that connects farmers with collection agents and recycling facilities, creating a transparent and efficient agricultural waste management ecosystem. The platform leverages modern web technologies and cloud infrastructure to deliver a real-time, responsive, and scalable solution that can operate across diverse geographical regions.

## 1.1 Aim of the Project

The aim of AgriTrace is to develop a comprehensive web-based platform for tracking, managing, and recycling agricultural waste. The platform facilitates the connection between farmers, waste collection agents, and recycling administrators through a role-based system that ensures efficient workflow management from waste generation to recycling completion.

The specific aims include:

- Provide farmers with a digital tool to report and list agricultural waste for collection.
- Enable collection agents to discover, claim, and manage waste pickup tasks efficiently.
- Empower administrators to oversee the entire waste management pipeline with analytics and user management capabilities.
- Track the environmental impact through carbon credit calculations.
- Incentivize participation through a gamification and rewards system.
- Integrate secure payment processing for waste transactions.

## 1.2 Motivation

The motivation behind AgriTrace stems from several critical factors:

1. **Environmental Crisis:** Open burning of agricultural waste contributes to approximately 25% of air pollution in northern India during October–November. A digital platform can help divert waste from burning to recycling.

2. **Economic Opportunity:** Agricultural waste has significant economic value when processed correctly. Rice husk, wheat stubble, and sugarcane bagasse can be converted into biofuels, compost, and industrial raw materials.

3. **Technology Gap:** While India has made strides in digital agriculture, there is a notable absence of platforms specifically addressing waste logistics, tracking, and recycling coordination.

4. **Policy Support:** The Indian government's push for sustainable agriculture and programs like the National Policy for Management of Crop Residues provide a supportive policy environment for technology-driven solutions.

5. **Carbon Credit Market:** The growing carbon credit ecosystem offers financial incentives for verifiable waste diversion, which AgriTrace can quantify and track.

## 1.3 Technology Overview

AgriTrace is built using a modern, production-grade technology stack carefully selected to deliver reliability, scalability, and excellent developer experience. The following is a brief overview of the core technologies used:

- **Next.js 15.5:** A React-based framework providing server-side rendering (SSR), static site generation (SSG), API routes, and file-based routing. Next.js enables building full-stack web applications within a single codebase, eliminating the need for a separate backend server. It supports middleware, incremental static regeneration, and optimized image handling out of the box.

- **TypeScript 5.0:** A statically typed superset of JavaScript that provides compile-time type checking, enhanced IDE support, and improved code maintainability. TypeScript's type system catches errors during development rather than at runtime, significantly reducing bugs in production.

- **Firebase:** Google's Backend-as-a-Service (BaaS) platform, providing:
  - *Cloud Firestore* – A NoSQL document database with real-time synchronization capabilities and offline support.
  - *Firebase Authentication* – Managed identity service supporting email/password, OAuth providers, and email verification.
  - *Firebase Storage* – Cloud-based file storage for waste report photos and delivery proofs.

- **Tailwind CSS:** A utility-first CSS framework that enables rapid UI development through composable utility classes. Combined with a custom design system, Tailwind CSS ensures visual consistency across all components.

- **Radix UI:** An open-source component library providing unstyled, accessible primitives for building high-quality design systems. Radix UI handles complex interactions such as dialogs, dropdowns, and tooltips with proper ARIA attributes.
- **Razorpay:** India's leading payment gateway, integrated for processing waste transactions. Razorpay supports UPI, credit/debit cards, net banking, and wallet payments with PCI DSS Level 1 compliance.
- **Google Genkit AI:** A framework for building AI-powered features. AgriTrace uses Genkit for intelligent waste classification, helping farmers identify waste types through AI-assisted categorization.
- **Cloudinary:** A cloud-based media management service used for uploading, storing, and optimizing waste report photographs with automatic image compression and format conversion.

## 1.4 Organization of the Report

This report is organized into fourteen chapters, each covering a specific aspect of the project:

- **Chapter 1 – Introduction:** Provides the background, aim, motivation, and technology overview of the AgriTrace project.
- **Chapter 2 – Problem Statement:** Defines the problem being addressed with supporting statistics and analysis of existing solutions.
- **Chapter 3 – Literature Survey:** Reviews existing research and systems related to agricultural waste management, digital platforms, and the technologies used.
- **Chapter 4 – Objectives and Scope:** Lists the specific objectives and defines the boundaries and scope of the project.
- **Chapter 5 – Requirement Analysis:** Documents the hardware and software requirements along with functional and non-functional requirements.
- **Chapter 6 – Methodology:** Describes the Agile Scrum development methodology adopted, including sprint planning and development tools.
- **Chapter 7 – System Design and Modeling:** Presents the system's design through Gantt charts, data flow diagrams, use case diagrams, activity diagrams, ER diagrams, and other UML models.
- **Chapter 8 – Coding:** Contains the key source code modules with detailed explanations of backend services, authentication, location services, notifications, payments, and UI components.

- **Chapter 9 – Output:** Showcases the application's user interface with screenshots of all major screens.
- **Chapter 10 – Testing:** Documents the test cases, testing methodologies, and results.
- **Chapter 11 – Cost Estimation:** Provides cost analysis using COCOMO model and itemized project expenditures.
- **Chapter 12 – Applications and Future Enhancements:** Discusses real-world applications and planned improvements.
- **Chapter 13 – Conclusion:** Summarizes the achievements, lessons learned, and closing remarks.
- **Chapter 14 – Bibliography:** Lists all references and sources cited in this report.